

Progress Report: High-Frequency Helmholtz with PINNs

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May 4, 2026

Objective: Solve high-frequency Helmholtz equation using domain decomposition.

Approach:

- Domain split into subdomains
- Independent PINNs per subdomain
- Interface continuity constraints (Hard and soft)

Observations:

- Improved convergence compared to vanilla PINN
- Better handling of local oscillations
- Reduced optimization instability

Conclusion: XPINNs mitigate spectral bias by reducing effective frequency per subdomain.

Highly Non-Convex Loss Landscape in Logarithmic Space

Idea Explored:

- Multiplicative Fourier features
- Log-space transformation:

$$\prod_j \phi_j(x) \rightarrow \exp \left(\sum_j \log \phi_j(x) \right)$$

Observed Issue:

- Extremely bumpy loss surface
- Sharp curvature variations
- Multiple flat valleys and saddle points
- Poor convergence even for moderate k

Reason:

- Log-exp introduces exponential sensitivity
- Ill-conditioned optimization landscape

Key Insight: Representation may be expressive, but optimization becomes unstable due to phase non-identifiability.

Ideas to be followed further:

- Incorporating CNNs? How?